243. The number $6n^2 + 6n$ for natural number n is always divisible by **(M.A.T., 2007)**
(*a*) 6 only (*b*) 6 and 12
(*c*) 12 only (*d*) 18 only

244. The difference of a number consisting of two digits and the number formed by interchanging the digits is always divisible by
(*a*) 5 (*b*) 7
(*c*) 9 (*d*) 11

245. The sum of a number consisting of two digits and the number formed by interchanging the digits is always divisible by
(*a*) 7 (*b*) 9
(*c*) 10 (*d*) 11

246. The largest natural number, which exactly divides the product of any four consecutive natural numbers, is **(S.S.C., 2007)**
(*a*) 6 (*b*) 12
(*c*) 24 (*d*) 120

247. If n is a whole number greater than 1, then $n^2 (n^2 - 1)$ is always divisible by
(*a*) 8 (*b*) 10
(*c*) 12 (*d*) 16

248. If n is any odd number greater than 1, then $n (n^2 - 1)$ is
(*a*) divisible by 24 always (*b*) divisible by 48 always
(*c*) divisible by 96 always (*d*) None of these

249. The sum of the digits of a 3-digit number is subtracted from the number. The resulting number is always
(*a*) not divisible by 9 (*b*) divisible by 9
(*c*) not divisible by 6 (*d*) divisible by 6

250. A number is multiplied by 11 and 11 is added to the product. If the resulting number is divisible by 13, the smallest original number is
(*a*) 12 (*b*) 22
(*c*) 26 (*d*) 53

251. The product of any three consecutive natural numbers is always divisible by
(*a*) 3 (*b*) 6
(*c*) 9 (*d*) 15

252. The sum of three consecutive odd numbers is always divisible by
I. 2 II. 3
III. 5 IV. 6
(*a*) Only I (*b*) Only II
(*c*) Only I and II (*d*) Only I and III

253. The greatest number by which the product of three consecutive multiples of 3 is always divisible is
(*a*) 54 (*b*) 81
(*c*) 162 (*d*) 243

254. If p is a prime number greater than 3, then $(p^2 - 1)$ is always divisible by
(*a*) 6 but not 12 (*b*) 12 but not 24
(*c*) 24 (*d*) None of these

255. The difference between the squares of two consecutive odd integers is always divisible by
(*a*) 3 (*b*) 6
(*c*) 7 (*d*) 8

256. A 4–digit number is formed by repeating a 2–digit number such as 2525, 3232 etc. Any number of this form is exactly divisible by **(S.S.C., 2005, 2010)**
(*a*) 7 (*b*) 11
(*c*) 13
(*d*) Smallest 3–digit prime number

257. A 6-digit number is formed by repeating a 3-digit number; for example, 256256 or 678678 etc. Any number of this form is always exactly divisible by
(*a*) 7 only (*b*) 11 only
(*c*) 13 only (*d*) 1001

258. The sum of the digits of a natural number $(10^n - 1)$ is 4707, where n is a natural number. The value of n is **(Hotel Management, 2010)**
(*a*) 477 (*b*) 523
(*c*) 532 (*d*) 704

259. $(x^n - a^n)$ is divisible by $(x - a)$
(*a*) for all values of n
(*b*) only for even values of n
(*c*) only for odd values of n
(*d*) only for prime values of n

260. Which one of the following is the number by which the product of 8 consecutive integers is divisible?
(*a*) 4 ! (*b*) 6 !
(*c*) 7 ! (*d*) 8 !
(*e*) All of these

261. Consider the following statements:
For any positive integer n, the number $10^n - 1$ is divisible by
(1) 9 for n = odd only (2) 9 for n = even only
(3) 11 for n = odd only (4) 11 for n = even only
Which of the above statements are correct?
(*a*) 1 and 3 (*b*) 2 and 3
(*c*) 1 and 4 (*d*) 2 and 4

262. If n is any positive integer, $3^{4n} - 4^{3n}$ is always divisible by
(*a*) 7 (*b*) 12
(*c*) 17 (*d*) 145

263. If the square of an odd natural number is divided by 8, then the remainder will be
(*a*) 1 (*b*) 2
(*c*) 3 (*d*) 4

264. The largest number that exactly divides each number of the sequence $1^5 - 1, 2^5 - 2, 3^5 - 3, ..., n^5 - n, ...$ is
(*a*) 1 (*b*) 15
(*c*) 30 (*d*) 120

265. The difference of the squares of two consecutive even integers is divisible by
(*a*) 3 (*b*) 4
(*c*) 6 (*d*) 7